Honghao Liu

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Education

Hong Kong University of Science and Technology **Sep. 2024 – present**
Ph.D. in Artificial Intelligence (Guangzhou campus), GPA: 4.2/4.3, Supervisor: Lionel Ni *Guangzhou, Guangdong*

Hong Kong University of Science and Technology **Sep. 2020 – Sep. 2022**
MPhil. in Data Science and Analytics, GPA: 3.7/4.3, Supervisors: Qiong Luo & Yang Xiang *Hong Kong*

Wuhan University of Technology **Sep. 2016 – Jun. 2020**
Bachelor of Engineering in Computer Science GPA: 88.6/100 *Wuhan, Hubei*

Publications

[1] **H. Liu**, C. Xu, X. Jiang, C. Yang, S. Yin, Z. Ma, L. Ni and J. Guo, *Conflicts Make Large Reasoning Models Vulnerable to Attacks*, In Findings of the ACL: **ACL 2026**, pages 9509–9531, San Diego, California, USA.
[2] C. Yang, X. Lin, X. Wu, C. Xu, X. Jiang, **H. Liu**, H. Xiong, J. Guo *Select2reason: Efficient instruction-tuning data selection for long-cot reasoning*, In Findings of the ACL: **ACL 2026**, pages 6656–6671, San Diego, California, USA.
[3] **H. Liu**, X. Jiang, C. Xu, C. Yang, Y. Cheng, L. Ni and J. Guo, *Continual Pretraining on Encrypted Synthetic Data for Privacy-Preserving LLMs*, In Findings of the ACL: **EACL 2026**, pages 408–425, Rabat, Morocco.
[4] J. Gu, X. Jiang, Z. Shi, H. Tan, X. Zhai, C. Xu, W. Li, Y. Shen, S. Ma, **H. Liu**, S. Wang, K. Zhang, Y. Wang, W. Gao, L. Ni and J. Guo, *A Survey on LLM-as-a-Judge*, **The Innovation**, 2024, doi: 10.1016/j.xinn.2025.101253.
[5] **H. Liu**, K. Li, Q. Zhou, Q. Wang, J. Sun, Z. Huang and J. Li, *Decoding multi-talker speech from EEG signals*, **IJCAI Workshop on Human Brain and Artificial Intelligence** (abstract), Jeju, South Korea, 2024.
[6] **H. Liu**, Q. Luo and F. Wang, *Efficient Radio Interferometric Imaging on the GPU*, **Proceedings of IEEE 18th International Conference on e-Science**, Salt Lake City, UT, USA, 2022, doi: 10.1109/eScience55777.2022.00024.

Research Experience

Hypothesis Ranking with Human–LLM Co-Evaluation and Wet-Lab Validation **Sep. 2026 – present**
Research intern, Harvard Medical School *Remote/Boston*
- Design the opencode plugin with agents for hypothesis generation and evaluation framework.

Automatic Agent Swarm for Future Prediction **June 2026 – present**
Research assistant, NLP Lab, The University of Hong Kong *Hong Kong*
- Design automatic agent generation and coordination process; Implement the opencode plugin as harness to perform the future prediction.

Multi-talker Speech Decoding from the Neural Activity **Jan. 2023 - May 2024**
Research Assistant, Comp. Linguistics, The City University of Hong Kong *Hong Kong*
- Collecting the EEG and fMRI data; Preprocessed fMRI data with the fMRIPrep and QSIPrep; Designed the multi-speaker decoding workflow, achieved BLEU-1 scores of 21% under both single- and multi- talker conditions.

Efficient Radio Interferometric Imaging on the GPU **Oct. 2020 – Sep. 2022**
The Hong Kong University of Science and Technology *Hong Kong*
- Designed and implemented and designed the non-uniform Fourier transform (NUFFT) in CUDA, parallelized the gridding and degridding methods; Optimized the memory access pattern and throughput and loading balance;

Work Experience

International Digital Economy Academy, Research Intern **Sep. 2024 – present**
AI Agent & Trustworthy AI *Shenzhen, Guangdong*
- FlowHarm: a benchmark for measuring safety of agentic workflow generation and Everything will follow: generalize the workflow generation framework by enabling autonomous, self-refining to adapt to diverse, real-world tasks.
- Proposed an entity-based method that synthesizes high-quality encrypted data to preserve personally identifiable information (PII) while enabling LLMs to securely encode new knowledge.

DJI, Research Intern **Jul. 2022 – Sep. 2022**
Accelerating perception algorithm for autonomous vehicles *Shengzhen, Guangdong*
- Implemented utility functions for the perception algorithms on the GPUs using CUDA C; Coordinated with the high-performance computing and perception teams; Applied this work to the listed Kiwi vehicles.

Additional Information

Skills: python, C++, CUDA C, MPI, Linux, Git, Multi-threading, LaTeX, solidity
Language: Mandarin (native), English (fluent)

Awards & Prizes

- Hong Kong University of Science and Technology Scholarship
- Wuhan University of Technology Scholarship

Coursework

- Parallel Programming (Teaching assistant)
- Advanced Algorithmic Techniques
- Computational Neuroscience (Summer school)
- Large Language Models: Theory and Practice